

Module 3: Complete Implementation Analysis Report.

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CS4412-DataMining.

1. *Implementation Status.*

The end-to-end analysis pipeline is fully operational. All the preprocessing, feature engineering, and mining stages required produce results. A summary of the results is presented in the subsequent table.

Component	Status	Key Output Metric
Imports, setup, and data loading.	Complete	All imported methods and dataset are present within the notebook and work as intended.
Data cleaning and preprocessing.	Complete	Negative values, null values, and cancellations within the dataset have been removed. The dataset was also restricted to working with invoices from UK customers which numbered 485,123 rows.
Association rules – Apriori	Complete	This resulted in 177 distinct rules, when the minimum support was set to 0.01, and a minimum confidence of 0.80.
Association rules – FP Growth	Complete	With equal min_support and min_confidence as Apriori, the same number of rules are encountered, 177.
RFM feature engineering	Complete	This process generates Recency, Frequency, and Monetary features for each customers giving them interpretable behaviors.
K – Means Clustering (k=4)	Complete	A silhouette score of 0.595
Hierarchical Clustering	Complete	A silhouette score of 0.606
Decision Tree – Classification	Complete	High average weighted macro f1
Gaussian Naïve Bayes- Classification	Complete	High average weighted macro f1

Seasonal Analysis	Complete	Transactions peak during autumn season.
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2. Analytical choices and Algorithm Comparisons.

- **The Apriori algorithm vs FP- growth.**

The two algorithms were run at identical thresholds where the min_support = 0.01, and a min_confidence = 0.80. They were run on same basket matrix which contained 18,019 invoices and 4,007 unique products. The matrix was configured to be sparse to ensure that enough memory could be allocated during its construction. A denser matrix caused the program to terminate prematurely. The algorithms were also tested on a min_support = 0.02 where only five rules were generated.

Metric	Apriori	FP-Growth.
Frequent itemset	Same	Same
Number of rules generated	177	177
Time taken	~5.6s	~6.6s

Note: The data in the table above is taken from the Apriori vs FP-Growth runtime graph.

Looking at the metric table, it would be advantageous to use the Apriori algorithm on future runs. The similarity also highlights the fact that no bias is introduced regardless of the algorithm chosen.

- **RFM feature engineering.**

This was performed to reduce each customer to a three-feature object to allow for standardization and uniformity. This resulted in 3,920 respective customers up until the last transaction date of 12-10-2011

- **K- Means vs Hierarchical Clustering.**

To begin, both the K-Means and Hierarchical Clustering algorithms were run at a k=4 and on standardized RFM features. This K =4 value was selected to ensure that a reasonable number of actionable segments are present, and this value lies at the elbow on the Elbow Method Graph.

The hierarchical clustering method corroborates the findings of the K-means algorithm showing that these clusters are naturally occurring within the data.

- **Classification – The decision tree and Gaussian Naïve Bayes.**

As there were no prior existing group labels, the k-means cluster assignments were used as class labels. These were: At risk/Lapsed customers, New/Occasional customers, Loyal Regulars, and Champions. Both algorithms used here were implemented to highlight and identify which RFM feature/s distinguish each segment.

For the decision tree, a max depth of four was chosen to maximize interpretability. The resulting importance levels were the following. The Monetary feature (importance = 0.368) served as the strongest splitter of the groups of customers, which should be the case since the dataset is a retail dataset. Followed by Recency = 0.331 and finally, Frequency of 0.301. From the respective confusion Matrices for Decision Tree and Naïve Bayes, it can be interpreted that both clearly identify the Champions but have some difficulty differentiating explicitly New/Occasional customers from those who are At Risk.

- **Anomaly Detection – DBSCAN vs Local Outlier Factor.**

After running both the DBSCAN and Local Outlier Factor algorithms, a small number of customers were identified to behave like true anomalies. The DBSCAN algorithm plotted about 52 noise points which accounted for 1.3% of the data. This indicates that the clusters present within the data are clean with relatively few outliers. On the other hand, the LOF marked about 5% of the customers as abnormal when compared to their respective neighbors. The two algorithms both highlight about 38 transactions as anomalous. Looking at the distribution of anomaly counts it can be inferred that anomalies exist across all customer segments and mostly occur within the At-Risk and New/Occasional customer segments. The graph showing the LOF anomaly score in PCA space highlighted that customers who had unusual RFM profiles were likely to be anomalous. The boxplot of anomalies vs normal monetary spend between customers also highlights that customers with a high Monetary value compared to normal customers are often anomalies.

- **Seasonal Analysis.**

Transactions are mapped to the seasons present in the year and then checked against the customer segments. First the revenue for each customer per season is calculated along with the number of invoices that occur within said seasons. The total revenue by season and customer segment graph shows how each customer segment influences the total revenue the store generated from valid purchases.

3. Results, Findings and Relation to Discovery Questions.

- **Results and Findings.**

- Association rules led to the identification of 3 main product families which are shown in the table below.

Product Family	Rules	Average Confidence	Average Lift	Interpretation.
Herb Markers	81	0.88	71.24	These are almost always bought by customers as a set. Thus, it would be practical to bundle these

				items together.
Jumbo Bag Collection (red retrospot and others)	36	0.84	9.5	
Regency Teacup Set(pink/green/roses)	20	0.87	26.8	These cups can be displayed together

- II. After RFM and K-Means analysis there were four customer segments identified. These are displayed in the table below.

Segment	Percent of customers.	Number of customers
Champions	0.1	3
At-Risk/ Lapsed	25.2	989
Loyal Regulars	1.3	52
New/Occasional	73.4	2876

- III. After performing seasonal analysis, all customer segments are at their peak during the autumn season as there are no At-Risk customers as seen in the heatmap. An uptick in the number of new customers occurs as well as those in the other segments. The monthly transaction volume graph shows that sales see an overall uptick in the months of September to November.

- **Relation to discovery questions.**

Discovery Question	Techniques	Answer
What products are frequently purchased together?	Apriori and FP-Growth	Three co-purchasing families are present as can be inferred from the association rules.
What natural customer segments exist?	RFM engineering and K-means	Four customer segments are present within the geo carved subset.
What distinct purchasing patterns emerge among customer segments by season?	Seasonal Analysis.	There is a universal peak purchasing behavior among all customer segments during the autumn season with champions accounting for 50% of purchases within said season.